

Srivilliputhur Municipality: Data Sources for an AI-Based Plastic Waste and Drainage Blockage/Overflow Risk Prediction Prototype

Scope: publicly verifiable sources identified for Srivilliputhur Municipality, Virudhunagar District, Tamil Nadu. This report preserves uncertainty: unavailable values are explicitly marked "Not Available." Sources in this report use the citation IDs and URLs shown in the source register.

Dataset classification

VERIFIED_PUBLIC = a currently accessible public/government/tender/news source. HISTORICAL_PUBLIC = older report/statistic useful for baseline analysis. FIELD_DATA_REQUIRED = student team must collect it. NOT_AVAILABLE = no reliable, public ward/street-level record identified.

Part	Dataset / finding	Classification	Primary source
1	Aggregate storm-water drainage network statistics; no public complete ward/street asset register.	HISTORICAL_PUBLIC	SFD Lite Report; municipal background source register
2	IMD rainfall portals and IMD daily weather reports; town-specific historic station export not verified in this search.	VERIFIED_PUBLIC	IMD rainfall information/statistics portals
3	News and sanitation-report evidence of heavy rain/flooding; ward-level incident log not found.	HISTORICAL_PUBLIC	SFD Lite Report; News18; Samayam Tamil
4	Municipal solid-waste information and town-level plastic/recyclable material notice; no drain-by-drain plastic quantity found.	VERIFIED_PUBLIC	Srivilliputhur Municipality portal
5	Town context, 33 wards, sanitation/drainage context.	HISTORICAL_PUBLIC	SFD Lite Report; Tamil Nadu Gazette
6	SFC 2023–24 SWD and culvert tender; individual BOQ measurements need tender documents.	VERIFIED_PUBLIC	Tamil Nadu eProcurement
7	Ward map, waterbody map, OpenStreetMap roads/waterways.	VERIFIED_PUBLIC	TN Urban Tree; Virudhunagar District; OSM
8	Condition, blockages, plastic accumulation, cleaning dates and local overflows require survey.	FIELD_DATA_REQUIRED	Field collection design in this report

Part 1 — Drainage master data

A complete 33-ward drain inventory was not found in the accessible public sources. The following entries are verified only to the level stated; missing dimensions must not be inferred.

Ward	Street / location	Drain or work description	Drain type	Length	Width	Depth	Culvert information	Road / municipal ID	Year	Source	Status
Not Available	Not Available	Municipal aggregate: storm-water drains along surfaced roads	Storm-water drain	92.4 km	Not Available	Not Available	Not Available	Not Available	Historical municipal statistic	Source 2 / Source 3	HISTORICAL_PUBLIC
Not Available	Not Available	Municipal aggregate: storm-water drains along unlined roads	Storm-water drain	8.7 km	Not Available	Not Available	Not Available	Not Available	Historical municipal statistic	Source 2 / Source 3	HISTORICAL_PUBLIC
Not Available	Not Available	Roads reported without drainage	Not Available	12.05 km	Not Available	Not Available	Not Available	Not Available	Historical municipal statistic	Source 2	HISTORICAL_PUBLIC
Multiple / Not Available	Srivilliputhur Municipality	Construction of SWD and culvert under SFC 2023–24, Pack 30	SWD + culvert	Not Available	Not Available	Not Available	Yes; details require tender BOQ	366/2023/E1	2023–24	Source 6	VERIFIED_PUBLIC
5	Ayyampatti Street	Construction of drainage culvert and pave-block road	Culvert	Not Available	Not Available	Not Available	Yes; dimensions Not Available	Not Available	2023 (search listing)	Source 7	VERIFIED_PUBLIC

All 33 wards: coverage status

Srivilliputhur has 33 wards according to the Tamil Nadu Gazette and public local-body listings. No accessible source found a complete drainage record for Ward 1 through Ward 33. Therefore every ward's street-level drain inventory remains FIELD_DATA_REQUIRED unless individual tender BOQs or municipal engineering registers are obtained.

Wards	Ward-level drain asset data	Public status	Action
1–33	Street, type, length, width, depth, culvert ID, construction year, condition	Not Available as a complete public dataset	Obtain municipal engineering register/RTI and validate through field mapping.

Part 2 — Rainfall data

IMD provides rainfall-information and rainfall-statistics portals, including district-wise daily/cumulative rainfall products. A directly downloadable, fully verified historical daily series specifically for the Srivilliputhur station was not retrieved during this research; obtain it from IMD or a formal data request before model training.

Date / period	Location / station	Rainfall (mm)	Duration	Source	Status
Long-term baseline	Srivilliputhur	811 mm annual average	Annual average; exact reference period should be checked in original report	Source 3	HISTORICAL_PUBLIC
16 Sep 2026 IMD daily report	Srivilliputhur, Virudhunagar district	3 cm (as reported by IMD; equals 30 mm)	Daily reporting period; verify report issue/date before data use	Source 4	VERIFIED_PUBLIC
07 Jul 2021	Srivilliputhur	Not Available	Reported as very heavy rain for about 1.5 hours	Source 8	HISTORICAL_PUBLIC
26 Oct 2024	Srivilliputhur and surrounding area	Not Available	Heavy rain reported	Source 9	HISTORICAL_PUBLIC
17 Dec 2024	Srivilliputhur temple area	Not Available	Rain/flood report	Source 10	HISTORICAL_PUBLIC

Part 3 — Flooding / waterlogging incidents

Date / year	Location / ward / street	Incident type	Cause stated by source	Rainfall stated	Source	Status
2019–2021	Multiple wards; exact wards Not Available	Open-drain pollution/possible drainage concern	Open drains described as carrying grey water and solid waste due to littering; no individual overflow event quantified	Not Available	Source 3	HISTORICAL_PUBLIC
07 Jul 2021	Srivilliputhur town	Heavy-rain impact; trees and electricity poles reportedly fell	Not Available	Not Available	Source 8	HISTORICAL_PUBLIC
17 Dec 2024	Temple area	Rainwater surrounding temple / flooding report	Not Available	Not Available	Source 10	HISTORICAL_PUBLIC
03 Jun 2026	Near Srivilliputhur	Flash-flood report	Not Available	Not Available	Source 11	VERIFIED_PUBLIC

Part 4 — Plastic and solid waste

No reliable public record was found that measures plastic accumulation or blockage at individual drains in Srivilliputhur. Municipal information does show that recyclable plastic and rubber were available at micro-composting/solid-waste facilities, but the notice does not provide a drain-level dataset.

Location	Ward	Waste / plastic issue	Evidence / description	Date / year	Source	Status
Srivilliputhur Municipality	Not Available	Recyclable plastic and rubber in municipal solid-waste system	Municipal site notice states that manure from micro-composting centres and recyclable plastic/rubber were in stock; quantity is Not Available in the search result	Current municipal portal page	Source 12	VERIFIED_PUBLIC
Multiple locations	Not Available	Solid waste in open drains	Sanitation report states open drains are commonly laden with household grey water and solid waste due to littering	2019–2021 report context	Source 3	HISTORICAL_PUBLIC
All individual drains	1–33	Plastic mass, debris category, blockage percentage	No reliable public dataset found	Not Available	Not Available	FIELD_DATA_REQUIRED

Part 5 — Municipal drainage and solid-waste context

Parameter	Value	Year / coverage	Official or primary source	Status
Municipal wards	33	2022 Gazette / current administrative context	Source 13; Source 14	VERIFIED_PUBLIC
Municipal area	5.718 km²	Report context	Source 3	HISTORICAL_PUBLIC
Storm-water drain length on surfaced roads	92.4 km	Historical municipal statistic	Source 2	HISTORICAL_PUBLIC
Storm-water drain length on unlined roads	8.7 km	Historical municipal statistic	Source 2	HISTORICAL_PUBLIC
Road length without drainage	12.05 km	Historical municipal statistic	Source 2	HISTORICAL_PUBLIC
Drain cleaning responsibility/frequency by ward	Not Available	Not Available	No verified public operations log identified	NOT_AVAILABLE
Open-drain/closed-drain segment count	Not Available	Not Available	No verified ward/segment asset register identified	NOT_AVAILABLE

Part 6 — Maintenance, construction and culvert works

Tender notices can establish that a proposed or procured work exists; they do not prove completion or current physical condition. Use the official tender BOQ, work order and completion certificate before treating project attributes as installed assets.

Year	Ward	Street / location	Work type	Length / quantity	Contractor / department	Tender / work ID	Source	Status
2023–24	Multiple / Not Available	Srivilliputhur Municipality	Construction of SWD and culvert under SFC 2023–24 Pack 30	Not Available	MAWS / Srivilliputhur Municipality; contractor Not Available	366/2023/E1	Source 6	VERIFIED_PUBLIC
2023	5	Ayyampatti Street	Drainage culvert and pave-block road	Not Available	MAWS; contractor Not Available	Not Available	Source 7	VERIFIED_PUBLIC
Not Available	1–33	Not Available	Routine desilting / drain cleaning record	Not Available	Not Available	Not Available	No public log found	NOT_AVAILABLE

Part 7 — GIS and location data

Dataset / map	Coverage	Information	Source	Download / API	Status
Srivilliputhur Ward Map	Municipality; visual coverage of wards	Ward-map image; GIS/shapefile availability Not Available	Source 15	Viewable municipal webpage	VERIFIED_PUBLIC
Virudhunagar district waterbody map	Srivilliputhur municipality/taluk map entry	Waterbodies and locations	Source 16	PDF map link from district site	VERIFIED_PUBLIC
OpenStreetMap	Town and surrounding region	Roads, water features and any community-mapped drains	Source 17; Source 18	OSM API/Overpass and editable map data; check feature completeness	VERIFIED_PUBLIC
Official municipal drain GIS layer	All wards	Drain alignment, dimensions, culverts, asset IDs	No source found	Not Available	NOT_AVAILABLE

DATA WE MUST COLLECT OURSELVES

The following variables are central to your ML model but cannot realistically be obtained as reliable, current, ward-level public data. Record the survey date/time and observer for every observation, and retain GPS-tagged photographs as evidence.

Variable	Why unavailable publicly	Field collection method	Suggested columns
Drain ID and geometry	No complete public asset inventory with segment dimensions	Assign a unique ID; capture GPS start/end; tape/laser-measure width, depth and segment length	Drain_ID, Ward, Street, Start_Lat, Start_Lon, End_Lat, End_Lon, Length_m, Width_m, Depth_m, Drain_Type, Date
Plastic accumulation	No drain-level plastic mass/volume dataset	Photograph and categorize debris; weigh safely collected representative samples if authorised	Drain_ID, Date, Plastic_kg, Plastic_Category, Debris_Score, Photo_ID, Observer_ID
Blockage percentage	No routine municipal blockage measurement	Estimate blocked cross-sectional area with a standard rubric; use repeated observer checks	Drain_ID, Date, Blockage_Percent, Sediment_Depth_cm, Water_Depth_cm, Observation_Score
Overflow event	No ward/drain event register found	Inspect after rainfall; collect timestamped photos and citizen/worker confirmation	Event_ID, Drain_ID, Date_Time, Overflow_YN, Overflow_Depth_cm, Photo_ID, Verification_Method
Cleaning history	No public cleaning log per drain	Request municipal logbooks and record direct observations/interviews; mark source reliability	Drain_ID, Last_Cleaning_Date, Cleaning_Type, Source_Type, Verified_YN, Days_Since_Cleaning
Historical local incidents	News rarely gives ward/street precision	Ward survey, municipal complaint-register request, and structured resident interviews	Incident_ID, Ward, Street, Date, Incident_Type, Severity, Source, GPS_Lat, GPS_Lon
Rainfall duration at site	Daily totals do not give local duration/intensity	Use IMD hourly data where available or install a calibrated rain gauge/logger	Date_Time, Station_ID, Rainfall_mm, Duration_min, Intensity_mm_hr, Source

ML variable availability

Proposed model input	Publicly usable now?	Recommended source / collection	Classification
Rainfall total	Partial	IMD district/station products; validate station and time resolution	VERIFIED_PUBLIC
Rainfall duration/intensity	Partial / Not Available locally	IMD hourly data if obtainable; otherwise local logger	FIELD_DATA_REQUIRED
Plastic accumulation	No	Field survey	FIELD_DATA_REQUIRED
Previous blockages	No	Field inspections / municipal complaint register	FIELD_DATA_REQUIRED
Previous overflow	Partial, non-spatial news only	Field event register; use news only as contextual evidence	FIELD_DATA_REQUIRED
Days since cleaning	No	Municipal logbook/field observations	FIELD_DATA_REQUIRED
Drain dimensions/type	Partial aggregate only	Field mapping plus BOQs/engineering register	FIELD_DATA_REQUIRED
Ward/location	Yes	Ward map, LGD/ward records, GPS survey	VERIFIED_PUBLIC
Historical waterlogging	Partial town-level reports	Create ward/street incident register through field/RTI	FIELD_DATA_REQUIRED

Source register (exact public URLs)

Use the source register below when verifying or downloading. Where a source is a tender listing, retrieve its original tender document/BOQ from the Tamil Nadu eProcurement portal before extracting physical measurements.

ID	Source / document	Exact URL
Source 1	Srivilliputhur Municipality — Public Disclosure	https://www.tnurbantree.tn.gov.in/srivilliputhur/public-disclosure/
Source 2	Srivilliputhur town drainage aggregate statistic (secondary compilation; corroborate against municipal records)	https://en.wikipedia.org/wiki/Srivilliputhur
Source 3	SFD Lite Report: Srivilliputhur, India (KARE / SuSanA PDF)	https://www.susana.org/_resources/documents/default/3-4228-7-1618294487.pdf
Source 4	India Meteorological Department — Rainfall Information	https://mausam.imd.gov.in/responsive/rainfallinformation.php
Source 5	India Meteorological Department — Rainfall Statistics	https://mausam.imd.gov.in/responsive/rainfall_statistics.php?PAGE=1
Source 6	Tamil Nadu eProcurement — Construction of SWD and Culvert in Srivilliputhur Municipality, SFC 2023–24	https://tntenders.gov.in/nicgep/app?component=\$DirectLink&page=FrontEndViewTender&service=direct&sp=S0ogkhgbRyi+5RcnVwJLlrQ==
Source 7	IndianTenders — drainage culvert and pave-block road, Ward 5 Ayyampatti Street (secondary tender listing)	https://www.indiantenders.in/tenderdetail/construction-drainage-culvert-and-pave-block-road-ward-no-5-ayyampatti-street-srivilliputhur-municipality-c365864/
Source 8	Samayam Tamil — heavy rain report, Srivilliputhur, 7 July 2021	https://tamil.samayam.com/tamil-nadu/virudhunagar/very-heavy-rain-at-srivilliputhur-for-one-and-half-hour-virudhunagar-power-cut-trees-fell-at-roads-around-20-eb-post-destroyed/articleshow/84198741.cms
Source 9	Samayam Tamil — heavy rain report, 26 October 2024	https://tamil.samayam.com/tamil-nadu/virudhunagar/heavy-rain-in-srivilliputhur-and-surrounding-areas/articleshow/114606791.cms
Source 10	News18 Tamil — temple flooding/rainwater report, 17 December 2024	https://tamil.news18.com/videos/tamil-nadu/srivilliputhur-temple-flood-rainwater-surrounds-the-temple-devotees-in-distress-rain-updates-tn-rain-1680120.html
Source 11	News18 Tamil/Valaioosai — flash-flood video listing, 3 June 2026	https://www.valaioosai.com/video/srivilliputhur-flash-flood-tn-rain-update-xXsjl1nO0Qs
Source 12	Srivilliputhur Municipality portal — solid-waste/recyclable plastic notice	https://www.tnurbantree.tn.gov.in/srivilliputhur/
Source 13	Tamil Nadu Government Gazette — ULB ward reservation document mentioning Srivilliputhur 33 wards	https://www.tnurbantree.tn.gov.in/tiruvannamalai/wp-content/uploads/sites/11/2022/01/ULB-Election-2022-Reservation-of-WardsChairman.pdf
Source 14	LocalBodyData — Srivilliputhur Municipality ward list (secondary reference)	https://localbodydata.com/municipality-srivilliputhur-252932
Source 15	Srivilliputhur Municipality Ward Map	https://www.tnurbantree.tn.gov.in/srivilliputhur/ward-map/
Source 16	Virudhunagar District — Waterbody maps	https://virudhunagar.nic.in/about-district/waterbody/
Source 17	OpenStreetMap	https://www.openstreetmap.org/
Source 18	OpenStreetMap Wiki — waterway tagging	https://wiki.openstreetmap.org/wiki/Key:waterway